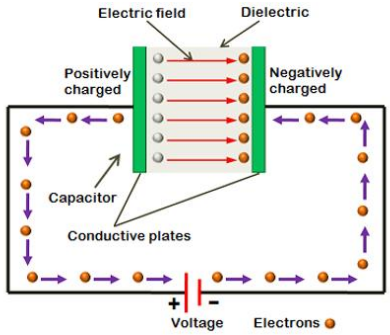
Chapter 40 – Capacitors and Dielectrics
Calculus Physics II

Capacitor – A capacitor consists of two parallel conducting plates that are charged with equal and opposite charges. The plates are separated by a distance d and the electric field established between the plates due to the charge separation is constant.

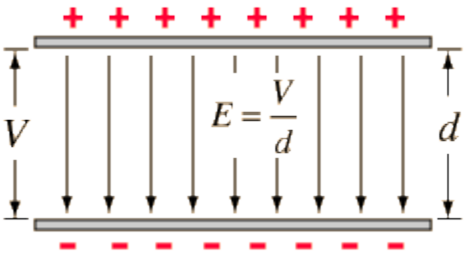


- One plate is charged with a net positive charge; this creates an attractive force on the opposite charge on the opposite plate, creating the same magnitude but opposite net negative charge on the opposite plate.
- To charge a capacitor, work must be done to move like charge to the same plate.
- The established electric field is where the energy is stored in the capacitor.
- Once the capacitor is charged (by power supply, battery etc.…) it can be removed from the charging device and the charge will stay on the capacitor, this is a convenient method of storing electrical energy.

Potential Difference For parallel plate Capacitor - The $\vec{E}$-field between two large parallel oppositely charged conducting plates has the same value everywhere between the plate (uniform electric field), we can therefore use this equation to relate the electric field $\vec{E}$ between the plates to the plate separation d and their potential difference V assuming vacuum/air is between the plates.

$$\Delta V = |\vec{E}|\,\Delta x = Ed$$

$$|\vec{E}| = \frac{\Delta V}{d}$$



$|\vec{E}|$ – Magnitude of Electric field strength.
d – plate separation
ΔV - potential between plates
Δx – displacement along electric field lines

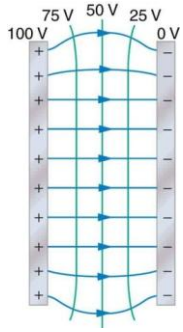
- The Electric field is constant, but the Electric potential is maximum near the positive plate and minimum near the negative plate.

Example: The capacitor shown has a potential difference of 100V with the equipotential lines shown at 25 V spacing. Determine the magnitude of the electric field between the plates if the spacing of the two plates was 10 cm.

$$E = \Delta V/d = (100V)/(0.1\ m) = 1{,}000\ V/m = 1{,}000\ N/C$$

Determine the potential 7.5 cm from the 0 V plate.

$$\Delta V = E\Delta x = (1{,}000\ N/C)(0.075\ m) = 75\ V$$



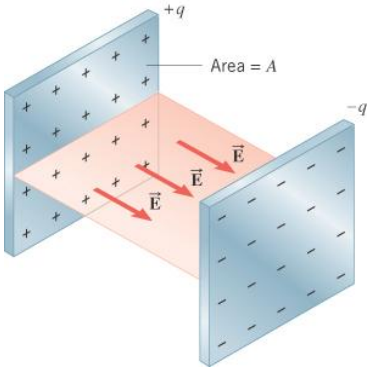
- The equipotential surfaces between the parallel plates are equally spaced because the electric field is uniform (expect for the edges), and the value of the electric field is the same over the inside of the capacitor.
- The electric field lines are shown going from the positive plate to the negative plate and the equipotential lines are perpendicular to the Electric field lines and parallel to the plates.

Surface charge density on the plates of a capacitor – The surface charge density (σ) on the plates of a parallel-plate capacitor is the charge (Q) per unit area (A), given by σ = Q/A. The surface charge density is directly related to the voltage (V) and plate separation (d) by

$$\sigma = \frac{\varepsilon_o V}{d}$$

This surface charge density is related to the electric field established between the plates, placing a Gaussian surface on one of the plates if $Q_{in} = \sigma A$ then

$$\Phi_E = \oint \vec{E} \bullet \hat{n} dA = \frac{Q_{in}}{\varepsilon_o} = \left|\vec{E}\right| A = \frac{\sigma A}{\varepsilon_o}$$



Solving for the electric field for a parallel plate capacitor

$$\left|\vec{E}\right| = \frac{\sigma}{\varepsilon_o}$$

Where sigma (σ) = charge per unit area = q/A = charge density.

ε_o is the permittivity of free space ($\varepsilon_o = 8.85 \times 10^{-12}\ C^2/Nm^2$)
q is the charge on one plate $q = \sigma A$
A is the area of one plate

<u>Example</u>: If two parallel conduction plates are separated by 10 mm, each having a surface area of 1 m^2 and air between the plates and a surface charge density $\sigma = 2.0 \times 10^{-8}\ C/m^2$ on each plate, what is the magnitude of the electric field between the plates? $\varepsilon_o = 8.85 \times 10^{-12}\ C^2/Nm^2$

$$E = \frac{\sigma}{\varepsilon_o} = \frac{2.0\ x\ 10^{-8}\ C/m^2}{(8.85\ x\ 10^{-12}\ C^2/Nm^2)} = 2{,}260\ N/C$$

Capacitance (C)- is the amount of charge a capacitor can hold at a given voltage, the capacitance of the device depends on the geometry and material between the plates of the capacitor.

$$C = \frac{q}{\Delta V}$$

"C" is the capacitance and has units of **Farads (F)** 1 F = 1 C/V
q is the net charge on one plate of the capacitor.
ΔV is the electric potential or voltage across the plates.

microFarads is μF (10^{-6} F), nF is nanoFarads (10^{-9} F) and picoFarads is pF (10^{-12} F)

The charge on a capacitor is given by the capacitance (C) of the capacitor and the potential (V) across the plates.

$$q = C\Delta V$$

For a parallel plate capacitor with vacuum (or air) between the plates, plate area (A) and plate separation (d) the capacitance (C) is given for vacuum or air by

$$C = \frac{\varepsilon_o A}{d} \quad C = \frac{q}{\Delta V} \quad C = \frac{q}{Ed} \quad E = \frac{q}{\varepsilon_o A} \quad E = \frac{q}{Cd} \quad \Delta V = Ed$$

- As the plate separation (d) decreased the capacitance increases.
- The larger the plates area (A) the larger the capacitance.

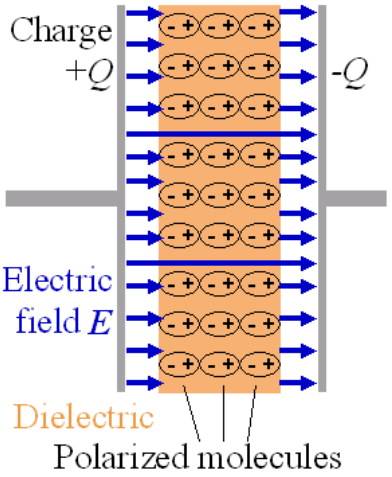
Permittivity (ε) is a fundamental electrical property of a material that describes how it interacts with and stores energy within an electric field. It essentially measures a material's ability to resist the formation of an electric field within it or, conversely, how easily an electric field can pass through it.

A material with higher permittivity has a greater opposition to the establishment of an electric field. This resistance arises from the polarization of the material's molecules when an external electric field is applied. These molecules (often dipoles, with positive and negative ends) align themselves in a way that generates an internal electric field opposing the external field. The greater the polarization, the higher the permittivity and the stronger the resistance to the external electric field.

Dielectric – Electrically insulating material that can be electrically polarized when the material is placed in an electric field.

If a dielectric is inserted between the plates of a capacitor, the capacitance can increase because of the way the dielectric modifies the electric field between the plates.

- More charge can be placed on a parallel plate capacitor when a dielectric is placed between the plates because of the polarization of the molecules in the dielectric creating an $\vec{E}$ field inside the dielectric in the opposite direction as the external $\vec{E}$ field.
- The amount of polarization of the material due to an external electric field is proportional to the material's dielectric constant and the permittivity.



<u>Dielectric Constant</u> (kappa - κ) - The Dielectric Constant is the ratio of the electric field without the dielectric (E_o) and with the dielectric (E).

$$\kappa = \frac{E_o}{E} \qquad \textit{and can also be given as} \qquad \kappa = \frac{V_o}{V}$$

- Kappa is a measure of the amount the electric field is reduced inside the dielectric when the dielectric is placed in an external electric field.
- E_o is the field with vacuum and E is the field with the dielectric.
- V_o is the potential across the capacitor with Vacuum and V is the potential with the dielectric.

Substance	Dielectric Constant (κ)
Vacuum	1
Air	1.00054
Teflon	2.1
Paper	3.3
Rubber	6.7
Water (pure)	80.4
Nylon	410

- The dielectric constant is also known as the relative permittivity and can be stated as the ratio of the permittivity of the material divided by the permittivity of free space.

$$\kappa = \frac{\varepsilon}{\varepsilon_o}$$

- The dielectric reduces the Electric field between the plates thereby reducing the potential across the plates.
- By filling the space between the plates of a capacitor with a dielectric, the capacitance of that capacitor is increased by the factor of the dielectric constant.

$C = \kappa C_o$ where C_o is the capacitance with no dielectric.

Charge and Capacitance of Parallel Plate Capacitor - The capacitance of a capacitor is affected by the geometry of the plates and the dielectric constant of the material between the plates and this determines the amount of charge the capacitor can hold.

$$C = \kappa\left[\frac{\epsilon_o A}{d}\right] \; and \;\; q = C\Delta V$$

$$q = \kappa\left[\frac{\varepsilon_o A \Delta V}{d}\right]$$

A – area of plates
ΔV – Voltage between the plates or V
d – separation of plates
κ – Dielectric constant
ε_o - Permittivity of free space ($\varepsilon_o = 8.85 \times 10^{-12}\ C^2/Nm^2$)

► Inserting a dielectric between the plates reduces the electric field between the plates, more charge can be placed on the plates at a given voltage thereby increasing the capacitance of the capacitor.

Example: A capacitor is made with two conducting plates that have an area of 1.5 m^2 and are separated by 2 mm with the space between filled with air, the capacitor is hooked up to a power supply that keeps the capacitor's potential at 12 V.

a) What is the capacitance of the capacitor?

$C = \varepsilon_o A/d = (8.85 \times 10^{-12}\ C^2/Nm^2)(1.5\ m^2)/(2 \times 10^{-3}\ m) = 6.6 \times 10^{-9}$ F or 6.6 nF

b) What is the charge on each plate of the capacitor if a voltage of 12 V is applied to this capacitor?

$C = q/\Delta V$ therefore $q = \Delta VC = 7.9 \times 10^{-8}$ C

c) What is the magnitude to the electric field between the plates?

$E = q/Cd = 6{,}000$ V/m

d) If instead of air a 2 mm thick piece of nylon (κ = 410) was placed between the plates keeping the potential at 12 V, what would be the new capacitance, charge and electric field?

$C = \kappa C_o = 410(6.6 \times 10^{-9}\ F) = 2.7 \times 10^{-6}$ F (410 times more)

$q = \Delta VC = 3.25 \times 10^{-5}$ C (410 times more)

The power supply had to add charge to the capacitor to keep it at 12 V!

E = q/Cd = 6,000 V/m (the same)

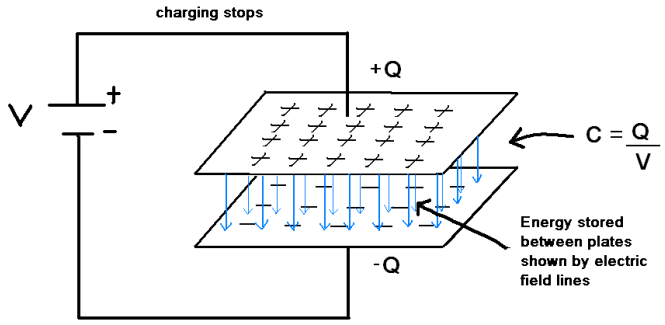
Because the dielectric reduced the electric field, the power supply added charge which brought the electric field back to its original value. If the capacitor was not hooked up to a source of charge (power supply), then the potential across the capacitor and the electric field would be reduced but the charge would be unchanged.

Energy Storage in a Capacitor - When a capacitor stores charge it also stores energy in the established electric field between the plates. It takes work to move the charge from one plate to the other plate (work is done by the power source; battery, power supply). This work/energy is stored potential energy in the capacitor and can be thought of as stored in the electric field between the plates.

$$\text{Stored Energy} = \tfrac{1}{2}CV^2$$

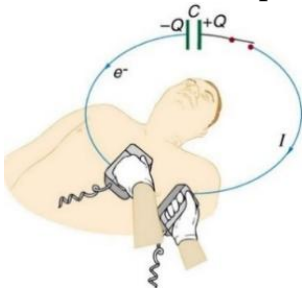
C – capacitance of the capacitor
V –potential difference or voltage across the plates of the capacitor
Energy stored in the electric field between the plates of the capacitor.

Stored Energy = $\tfrac{1}{2}CV^2$



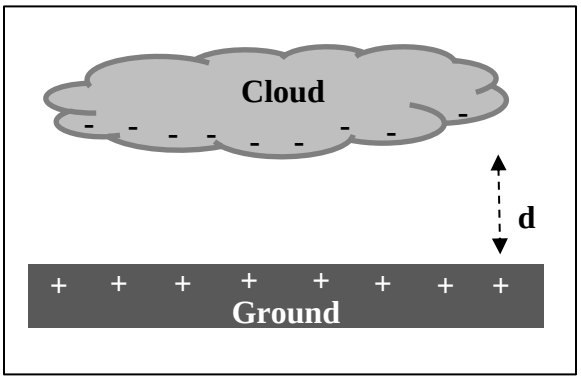
Example: A Defibrillator uses a charged capacitor that is charged to a high voltage to create the charge flow through the heart muscle that gets the heart going again. If the capacitor has a capacitance of 30 µF and is charged to 5,000 V, how much energy is stored in the capacitor?

Energy = $\tfrac{1}{2}CV^2 = \tfrac{1}{2}(30 \times 10^{-6}\ \text{F})(5000\ \text{V})^2 = 375\ \text{J}$



Example: When the bottom of a cloud becomes negatively charged it attracts the positive change from the ground, this can be thought of as a type of capacitor, with the cloud and ground being the parallel plates with air in between.

If it takes an electric field strength of 3×10^6 V/m to breakdown (conduct charge) the air between the cloud and ground, how much energy is stored in the cloud due to the charge right before the lightning is formed (before the air breaks down)? Assume the distance between the cloud and ground is about d = 1,000 m and the area of the cloud is approximately 8×10^7 m^2 (about 5 km radius)



$V = Ed = (3 \times 10^6 \text{ V/m})(1{,}000 \text{ m}) = 3 \times 10^9 \text{ V}$

$C = \varepsilon_o A/d = (8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2)(8 \times 10^7 \text{ m}^2)/(1{,}000 \text{ m})$ $C = 7.1 \times 10^{-7}$ F

Stored Energy = $\frac{1}{2}CV^2 = \frac{1}{2}(7.1 \times 10^{-7} \text{ F})(3 \times 10^9 \text{ V})^2$

Stored Energy = 3.2×10^{12} J

If the average house uses about 2,000 W of electrical power, if it were possible to collect and store this energy (unfortunately it is not), how long would it run the house?

Power = Energy/time Solving for time

time = Energy/Power = $(3.2 \times 10^{12} \text{ J})/2{,}000 \text{ W} = 1.6 \times 10^9$ s $\approx$ 50 years

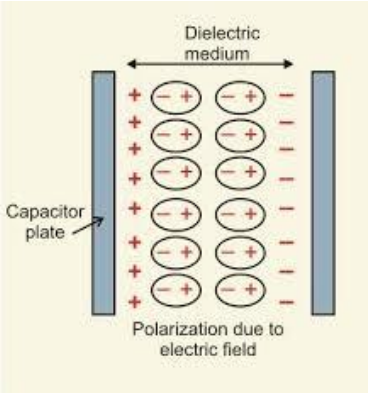
Safety Precautions using Capacitors – Capacitors can store large amounts of energy and this energy can be released very fast creating large amounts of power.

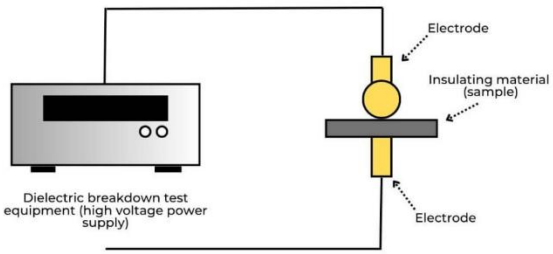
- Always discharge capacitors before working with them.
- Be very careful around old TV's, old Computer monitors even if they are unplugged and have been for a long period of time. They still can have a lethal charge!

Example of typical Capacitors



Dielectric Strength and Breakdown Voltage – When a dielectric is in an electric field the outer electrons in that dielectric material experience a force due to the electric field and the electrons, the atoms/molecules become polarized. If the electric field becomes large enough these electrons will be stripped off the molecules and free to move along the electric field, at this point an avalanche of electron's become dislocated and a current is established between the charge separation. This is known as the breakdown voltage and the properties of the material are destroyed along with the capacitor.





Break Down Voltage (V_{Bd}) – is the minimum voltage required to create a current though a material due to the electrons being dislocated from the atoms in the material.

Dielectric strength (DS) - of a material is the maximum electric field (V/m) that a material can experience before breakdown. The breakdown voltage of a dielectric is given by

$$V_{Bd} = (DS)d$$

DS is the dialectic strength of the material (in V/m)
d – is the thickness of the material measured along the electric field lines (in meters)

TABLE 26.1 Dielectric Constants and Dielectric Strengths of Various Materials at Room Temperature

Material	Dielectric Constant κ	Dielectric Strength[a] (V/m)
Air (dry)	1.000 59	3×10^6
Bakelite	4.9	24×10^6
Fused quartz	3.78	8×10^6
Neoprene rubber	6.7	12×10^6
Nylon	3.4	14×10^6
Paper	3.7	16×10^6
Polystyrene	2.56	24×10^6
Polyvinyl chloride	3.4	40×10^6
Porcelain	6	12×10^6
Pyrex glass	5.6	14×10^6
Silicone oil	2.5	15×10^6
Strontium titanate	233	8×10^6
Teflon	2.1	60×10^6
Vacuum	1.000 00	—
Water	80	—

[a] The dielectric strength equals the maximum electric field that can exist in a dielectric without electrical breakdown. Note that these values depend strongly on the presence of impurities and flaws in the materials.



Dielectric breakdown in air

Example: The space between the capacitor's plates is filled with dry air, the spacing of the plates is 0.5 mm, what is the maximum voltage the capacitor can have before breakdown?

$$V = (DS)d = (1 \times 10^{6}\ V/m)(0.5 \times 10^{-3} m) = 500\ V$$

What if the space was filled with nylon?

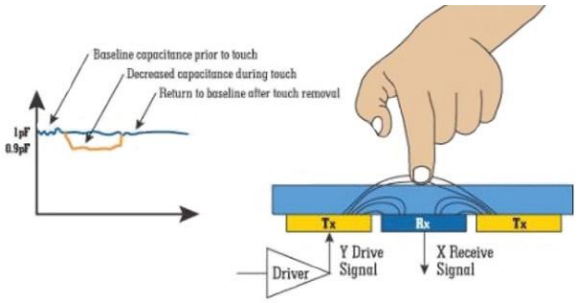
$$V = (DS)d = (14 \times 10^{6}\ V/m)(0.5 \times 10^{-3} m) = 7{,}000\ V$$

Applications for capacitors

- Flash on camera – A battery charges the capacitor and when the picture is taken the capacitors stored energy (charge) goes to the discharge tube that create the light.



- Touch screen on electronic devices – the finger changes the capacitance between conducing points on a grid that is built into the touch screen, the change in capacitance is detected by the electronics which gives the position of the touch.



- Defibrillator – capacitors are charged to a high voltage (potential) and the electrodes are placed so that the electoral discharge will go through the heart, this discharge can stimulate the heart to start beating again.



- They are also used in electric circuits as temporary energy storage, frequency filters among others circuit applications.